Lesson plan · for the teacher

Current, LEDs and resistors

IoT · Level 2 · Lesson 1

Time
40 min

Materials
Two AA batteries per group, in a holder if you have one, and tape; A torch bulb, or a red LED with a 100 Ω resistor; Three short wires, or strips of kitchen foil; Ten everyday objects: coins, keys, a pencil, a ruler, an eraser, foil

Learning goals

- Explain that current flows only round a closed loop, pushed by the battery's voltage.
- Light an LED the right way round, with a resistor to limit the current.
- Use Ohm's law to choose a resistor, and spot a short circuit.

Activity: Conductor hunt

1. Warm-up (5 min): Show a torch with a loose wire or a battery in backwards. Ask: “The batteries are new. Why won't it light?” Collect ideas on the board.
2. Teach (8 min): Draw the loop: battery, wire, bulb, and back to the battery. Explain voltage as the push and current as the flow. Show an LED's long and short legs, and why it needs a resistor.
3. Build (7 min): Each group tapes two AA batteries end to end, plus to minus, and connects the bulb (or the LED and its resistor) with wires, leaving a gap between two free ends. Touching the ends together should light it.
4. Test (15 min): Groups predict each object first, then touch both free ends to it and record “lights” or “stays dark” in a table. They copy their results into two columns on the board.
5. Wrap-up (5 min): Ask: why is a wire copper inside plastic? Exit ticket: one conductor, one insulator, and one safety rule.

Before class, build and test one tester yourself. Use batteries only. Keep button batteries away from young children, and don't let pupils hold a bare wire across a battery: it gets hot.

Discussion

- Which result surprised you, and why?
- Why does an LED need a resistor when a torch bulb doesn't?
- Where could a short circuit happen by accident at school or at home?

Exit ticket

- Name one conductor, one insulator, and one safety rule for circuits.

Try it in class tomorrow
Conductor hunt: Make a tester from two AA batteries, a torch bulb (or a red LED with a 100 Ω resistor) and three wires. Groups predict, then test ten everyday objects and sort them into conductors and insulators on the board. Finish by asking: why is a wire copper inside plastic?

Worksheet · for pupils

Current, LEDs and resistors

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Name _____ Class _____ Date _____

1 **Kemi wants her class's LED to be brighter, so she plans to take out its resistor. What should you tell her?**

- ◯ Go ahead: without the resistor it will be at its brightest
- ◯ Don't: it will burn out. Use a smaller resistor instead
- ◯ Don't: an LED without a resistor won't light at all

2 **A torch with brand-new batteries and a working bulb must light up.**

◯ True ◯ False

3 **Pupils touch each object across the gap in a battery-and-bulb tester. Which will light the bulb?**

☐ A steel key ☐ A plastic ruler ☐ A coin ☐ A rubber eraser ☐ A strip of kitchen foil ☐ A dry wooden stick

4 **Choose a resistor for a red LED on a 5 V USB supply.**

A red LED takes 2 V, so _____ volts are left for the resistor. For 15 mA, the resistor needs about _____ ohms.

Word bank: 3 · 200 · 5 · 330

5 **Match each part to its job in the circuit.**

Battery	• • • •	Opens or closes the loop
Wire	• • • •	Turns current into light, one way round
Resistor	• • • •	Limits the current
LED	• • • •	Carries current round the loop
Switch	• • • •	Gives the push (voltage)

✎ **Which everyday objects from your pupils' homes could you use to teach circuits? Where could a short circuit happen by accident at school?**